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Subject ACV

ISE 1.1

```
import cv2
import matplotlib.pyplot as plt
import urllib.request
import os

# Your Google Drive sharing URL
sharing_url =
"https://drive.google.com/file/d/17ion9uP9RXGvEcs5iPvIC0VzsbzDcZJJ/
view?usp=sharing"

# Extract the file ID from the sharing URL
# The ID is between '/d/' and '/view'
file_id = sharing_url.split('/d/')[1].split('/view')[0]

# Convert it to a direct download URL for Google Drive files
direct_download_url = f"https://drive.google.com/uc?
export=download&id={file_id}"

image_path = 'kitten.png'

# If a corrupted file (like an HTML webpage) already exists, let's
remove it first
if os.path.exists(image_path):
    # We check if the existing file is extremely small or not a real
image to prevent using a corrupted download
    img_test = cv2.imread(image_path)
    if img_test is None:
        print("Removing invalid/corrupted local file...")
        os.remove(image_path)

# Download the actual image using the direct download link
if not os.path.exists(image_path):
    print("Downloading image from Google Drive...")
    # Using user-agent header to avoid Google Drive download blocks
    opener = urllib.request.build_opener()
```

```
    opener.addheaders = [('User-agent', 'Mozilla/5.0')]
    urllib.request.install_opener(opener)
    try:
        urllib.request.urlretrieve(direct_download_url, image_path)
    except Exception as e:
        print(f"Could not download sample image: {e}")

# Load the color image
img = cv2.imread(image_path)

# Verify if the image was successfully loaded
if img is None:
    print(f"Error: Could not load image '{image_path}'. Please make
sure the file exists and is a valid image.")
else:
    # Convert color image to grayscale
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Display original and grayscale images
    plt.figure(figsize=(10, 5))

    plt.subplot(1, 2, 1)
    plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
    plt.title("Original Image")
    plt.axis("off")

    plt.subplot(1, 2, 2)
    plt.imshow(gray, cmap='gray')
    plt.title("Grayscale Image")
    plt.axis("off")

    plt.show()

    # Save the grayscale image
    cv2.imwrite('/content/grayscale.jpg', gray)
    print("Grayscale image saved successfully as
'/content/grayscale.jpg'.")
```

```
Removing invalid/corrupted local file...
Downloading image from Google Drive...
```

Original Image



Grayscale Image



```
Grayscale image saved successfully as '/content/grayscale.jpg'.
```